In JEE Advanced 2016, FIITJEE Students bag 36 in Top 100 AIR, 75 in Top 200 AIR, 183 in Top 500 AIR. 3541 Students from Long Term Classroom/ Integrated School Program & 4423 Students from All Programs have qualified in JEE Advanced, 2016

ALL INDIA TEST SERIES

FIITJEE

JEE(Advanced)-2017

ANSWERS, HINTS & SOLUTIONS

FULL TEST – II
(PAPER - 2)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	B	A	B
2.	D	C	B
3.	D	A	C
4.	C	B	D
5.	B	A	B
6.	D	B	B
7.	B	C	B
8.	A	B	D
9.	BCD	ABC	AD
10.	AD	ACD	ACD
11.	AC	ABC	BD
12.	ABC	AC	AD
13.	D	D	D
14.	C	B	A
15.	C	B	C
16.	A	C	A
17.	C	B	D
18.	A	D	D
1.	6	0	2
2.	5	4	4
3.	3	3	2
4.	2	6	7
5.	2	9	2

Physics

PART – I

SECTION – A

1. $\lambda=\frac{v}{f}=\frac{330}{500}=0.66m=\frac{4\lambda}{2n-1}\Rightarrow n=3$

2. $\Delta x=2d=\lambda,\ \text{Now, } U=fx=2fd$

3. $V_1=\frac{kQ}{\sqrt{2}b};V_2=\frac{kQ}{\sqrt{5}b}$

4. If another shell is kept upside down over it complete a sphere, net field should become zero

5. $F_1=F_B-\left(\pi a^2\right)\rho gh$
   $F_2=F_B-\left(\pi a^2\right)\rho g\left(2h\right)$

6. $\mu_2>\mu_1>\mu_3$
   $v_2<v_1<v_3$
   $\lambda_2<\lambda_1<\lambda_3$
   Frequency is independent of medium

7. B_w is magnetic field of wire
   $B_1=B_0+B_w$
   $B_2=B_0-B_w$
   $\Rightarrow B_0=\frac{B_1+B_2}{2}$

8. $C_V>\frac{fR}{2}$
   f is $=3$ for monoatomic but f is 5 for diatomic at normal temperature. But $f>5$ for diatomic gases. At high temperature energy of vibration also increases taken that into account C_V increases since molecules of monoatomic gas do not vibrate, its C_V remains same.

9. The loops is always attracted towards wire as the region part of loop getting attracted is experiencing stronger magnetic field

10. The magnetic field due to the outer loop is into the paper. The force on an element of the inner loop has the direction given by $\vec{ds}\times\vec{B}$ where $\vec{ds}$ is the element length in the direction of current. Thus the force is radially outward.
    Since the force is symmetric, i.e. radially outwards everywhere and the loops are concentric, there is no net force on the inner loop.

11. Equivalent diagram is as shown in P is moved 2 cm right them R_1=12, R_3=3

$\frac{R_1}{R_2}=\frac{R_2}{R_4}\left(\text{Hence wheat stone will be balanced}\right)$

If s is moved left $\frac{5}{3}cm$ then $R_4=\frac{20}{3}$ hence $\frac{R_1}{R_3}=\frac{R_2}{R_4}$ (hence wheat stone will be balanced)

12. $I\omega=\tau\Delta t$

Hence $\frac{1}{2}I\omega^2=\frac{\left(\tau\Delta t\right)^2}{2I}$

15. $f\propto\sqrt{T}$

So Δf increases by increasing T.

i.e. $f_2=f_1+3=443\,Hz$

16. $v=f\lambda$

$\lambda=\frac{340}{360}=0.94$

17. Pressure will increase, equally at all points

18. The compression will be proportional to g_{eff}

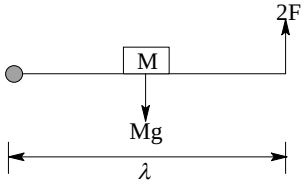
SECTION – C

1. By linear momentum conservation impulse (J) = mV.

By angular momentum conservation, angular impulse $=J\frac{\lambda}{2}=I\omega$

So $mv\frac{\lambda}{2}=I\omega\ or\ \omega=\frac{mv\lambda}{2I}=\frac{mv\lambda}{2\left(\frac{m\lambda^2}{12}\right)}=\frac{6v}{\lambda}=6\,rad/s$

2.



For equilibrium

$\left|Mg\frac{\lambda}{2}\right|=\left|2F\right|1$

$F=\frac{mg}{4}=\frac{100}{4}=25\,N$

Chemistry

PART – II

SECTION – A

1. $\log\frac{P_2}{P_1}=\frac{\Delta H}{2.303R}\left[\frac{1}{T_1}-\frac{1}{T_2}\right]$

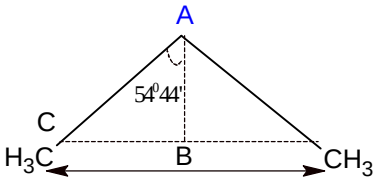
$\log\frac{P_2}{760}=\frac{9720}{2.303\times2}\left[\frac{1}{373}-\frac{1}{348}\right]$

$P_2=298\text{torr}$

$\frac{298-P_{sol.}}{298}=\frac{0.1}{0.1+\frac{1000}{18}}\approx\frac{0.1}{\frac{1000}{18}}$

$P_{sol}=297.46\text{torr}$

Lowering in VP=(298-297.46)=0.54torr

2. 

$\text{Sin}(54^0\,44')=\frac{CB}{AC}$

Terminal carbon carbon distance $=2\times CB$

$=2\times\ 1.54\ \times\text{Sin}(54^0\,44')$

$=2.51\ A^0$

3. 0.5.mol of CH_3COONa and 0.5mol H_2O is formed.

$\Delta T_f=\frac{0.5\times2}{20.5\times18\times10^{-3}}=5.04$

4. 100g H_2O_2 per hr.
(100/34)mol H_2O_2 per hr.
(100/34)mol $(NH_4)_2S_2O_8$ per hr.

$\frac{100\times2}{34}\text{mole e}^-\text{ per hr}$

i =157.68amp

current required=315 amp

5. $E_{cell}=E^0{}_{cell}-\frac{0.059}{1}\log Q$

$0=E^0{}_{cell}-\frac{0.059}{1}\log K$

$0=(x-0.799) - \frac{0.059}{1}\log(6\times10^{-8})$

$x = 0.37$

6. $SO_3^{2-} + 2H^+ \longrightarrow SO_2 + 2H^+$

7. A, B, D is having plane of symmetry.

8. Intercept=$E^0_{Cu^{2+}/Cu}$=0.34

$E_{Cu/Cu^{2+}}=E_{Cu/Cu^{2+}} - \frac{0.059}{2}\log[Cu^{2+}]$

$E_{Cu/Cu^{2+}} = -0.34 - \frac{0.059}{2}\log 0.1$

$E_{Cu/Cu^{2+}} = -0.3105V$

9. In isothermal process $\Delta U=0$,

10. $\frac{35.6}{24}$ mole Mg

$(\frac{35.6}{24}\times 2)$ mole of e^-

$(\frac{35.6}{24}\times 2\times 96500)$ coloumb

12. C_4H_{10},CH_3CH_2OH, NH_3 and I_2 have high BP so low vapour pressure.

(13 – 14).

$FeSO_4.7H_2O \xrightarrow{\text{heat}} FeSO_4(B)+H_2O$

$FeSO_4.7H_2O \xrightarrow{\text{strong heat}} Fe_2O_3(D)+SO_2(E)+SO_3(F)$

(15 – 16)

$H_2O_2+2KI+H_2SO_4 \longrightarrow K_2SO_4+I_2+2H_2O$

$I_2 + Na_2S_2O_3 \longrightarrow 2NaI + Na_2S_4O_6$

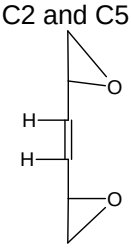
4mmol of hypo means 2mmole of I_2 and 2 mmole of H_2O_2.

so molarity=$\frac{2}{25}$, $normality = \frac{4}{25} = 0.16N$

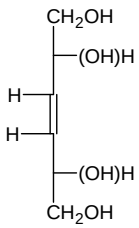
And volume strength=$\frac{2}{25}\times 11.2V=0.896V$

17. C2 and C5 should having opposite configuration for S to be meso.

18. C2 and C5 should having opposite configuration for S to be meso it is possible with D only.



(Q)



S,Meso(C2 and C5 have opposite configuration)

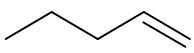
SECTION – C

1. Pyrolusite $\longrightarrow MnO_2$
   Iron Chromite $\longrightarrow FeO.Cr_2O_3$
   Siderite $\longrightarrow FeCO_3$
   Cassiterite $\longrightarrow SnO_2$
   Calamine $\longrightarrow ZnCO_3$
   Argentite $\longrightarrow Ag_2S$
   Lime stone $\longrightarrow CaCO_3$
   Chalcopyrite $\longrightarrow CuFeS_2$

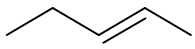
2. $NH_3, SO_2, Ca(OH)_2, NaCl$

3. I,ii,and V are correct

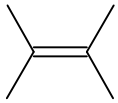
4. Sc^{3+} &Co^{3+} are diamagnetic Sol.



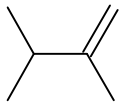
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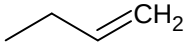
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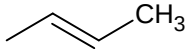
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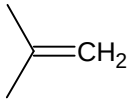
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(1)



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Mathematics

PART – III

SECTION – A

1. $a\left(\frac{1+\sqrt{5}}{2}\right)^9 + b\left(\frac{1+\sqrt{5}}{2}\right)^8 + 1 = 0$

$2^9 a + 2^8 b\left(\sqrt{5}-1\right) + \left(\sqrt{5}-1\right)^9 = 0$

$2a + b\left(\sqrt{5}-1\right) = 76 - 34\sqrt{5}$

$\Rightarrow \quad 2a - b = 76,\ b = -34 \Rightarrow a = 21$

2. $2ab - 5a + b = 55$

$\Rightarrow \quad (2a+1)\left(b - \frac{5}{2}\right) = 55 - \frac{5}{2}$

$\Rightarrow \quad (2a+1)(2b-5) = 105 = 3.5.7$

Total number of positive solutions $= 2 \times 2 \times 2 = 8$

Total number of solutions $= 16$.

3. $f_n = 2^n + 3^n$

$\Rightarrow \quad f_{n+1} - 2f_n = 3^n$ and $f_{n+2} - 2f_{n+1} = 3.3^n$

So option '*c*' is correct.

4. $P_n = \frac{3!.(3n-3)!.(n!)^3}{((n-1)!)^3 (3n)!} = \lim_{n\to\infty} \frac{6}{(3n)} \frac{n^3}{(3n-1)(3n-2)}$

$= \frac{2}{9}$

5. $\left(\frac{a+c}{ac} - 2\right) bx - 2y + 7b = 0$

$\Rightarrow \quad \left(\frac{2}{b} - 2\right) bx - 2y + 7b = 0$

$\Rightarrow \quad x - y - b\left(x - \frac{7}{2}\right) = 0$

Line passes through $\left(\frac{7}{2}, \frac{7}{2}\right)$ maximum distance from origin $\sqrt{\left(\frac{7}{2}\right)^2 + \left(\frac{7}{2}\right)^2} = \frac{7}{\sqrt{2}}$ and minimum is zero. $\left(0, \frac{7}{\sqrt{2}}\right]$

6. Let roots of the equation $x^3 - 13x^2 + 54x - 72 = 0$ are, a, b, c

$\sum a = 13$

$\sum ab = 54$

$abc = 72 \qquad S = \frac{13}{2}$

or $\quad x^3 - 13x^2 + 54x - 72 = (x-a)(x-b)(x-c)$

$$\left(\frac{13}{2}\right)^3 - 13\left(\frac{13}{2}\right)^2 + 54\left(\frac{13}{2}\right) - 72 = (S-a)(S-b)(S-c)$$

$$\frac{35}{8} = (S-a)(S-b)(S-c)$$

$$\Delta = \sqrt{S(S-a)(S-b)(S-c)}$$

$$= \frac{\sqrt{455}}{4}$$

7. Clearly α is repeated root of f(x)=0

$$\lim_{x\to\alpha} \frac{f(x)}{f'(x)}\left(\frac{f'(x)}{f(x)} - \left\{\frac{f'(x)}{f(x)}\right\}\right) \quad \Rightarrow \lim_{x\to\alpha}\left(1 - \frac{f(x)}{f'(x)}\left\{\frac{f'(x)}{f(x)}\right\}\right) = 1-0=1$$

$$\left(\lim_{x\to\alpha}\frac{f(x)}{f'(x)} = 0 \;\&\; \left\{\frac{f'(x)}{f(x)}\right\} \text{is bounded function}\right)$$

8. Clearly if $f(x) = a(x-\alpha)(x-\beta)$ then $g(x) = b(x-\alpha)(x-\beta)$

$\Rightarrow h(x) = k(x-\alpha)^2(x-\beta)^2 \qquad \Rightarrow h(x)h'(x) = 2k(x-\alpha)^3(x-\beta)^3(2x-\alpha-\beta)$

So distinct roots of $\frac{d}{dx}(h(x)h'(x)) = 0$ are 4.

9. $$p\left(\frac{x}{z}\right)^2 + \left(\frac{y}{z}\right)^2 + 3\left(\frac{x}{z}\right)\left(\frac{y}{z}\right) + \left(\frac{x}{z}\right) + 2\left(\frac{y}{z}\right) + q = 0$$

To factories make $\Delta = 0$

$$p\times 1\times q + 2\times\frac{1}{2}\times 1\times\frac{3}{2} - p\times 1^2 - 1\times\left(\frac{1}{2}\right)^2 - q\left(\frac{3}{2}\right)^2 = 0$$

and also for perpendicularity $p + q + 1 = 0$

$\Rightarrow \qquad 4p^2 - p - 14 = 0 \qquad (4p+7)\,(p-2) = 0$

$$p = 2,\ q = -3, \qquad p = \frac{-7}{4},\ q = \frac{3}{4}$$

10. Let $\frac{dy}{dx} = p$, we have

$$p^2 + y = xp$$

$\Rightarrow \qquad 2pp' + p = p + xp'$

$\Rightarrow \qquad 2p = x$ or $p' = 0$

which givens $x^2 = 4y$ or $y = cx + (2c+1)$

or $y - 1 = c(x+2)$

11. Fro more than one triplet satisfying all three equations we have

$$\begin{vmatrix} \frac{1}{a} & \frac{1}{b} & \frac{1}{c} \\ \frac{1}{\mathrm{b}} & \frac{1}{c} & \frac{1}{a} \\ \frac{1}{c} & \frac{1}{a} & \frac{1}{b} \end{vmatrix} = 0$$

$\Rightarrow \quad a = b = c$

$\Rightarrow \quad (x + y + z) = a$ satisfying the triplets

We get $a = 10, \quad p = 7 \quad \Rightarrow \quad \alpha^2 - 2\alpha - 8 = 0$

$\alpha = -2$ and 4.

12. If $f(x)$ is continuous then $\lim\limits_{k\to\infty} f(t_k) = f\left(\lim\limits_{k\to\infty}(t_k)\right) = f\left(\frac{1}{4}\right) = 1$ (Where $\lim\limits_{k\to\infty} t_k = \frac{1}{4}$).

Also as graph is smooth $f'\left(\frac{1}{4}\right) = 0$

13. $h(n) = \int_o^a 2 \,|\sin n\pi x|\, dx$

Let $\quad a = \frac{k}{n} + \alpha, \quad 0 \le \alpha < \frac{1}{n}, \quad k \in I$

$$= k\int_0^{1/n} 2\,|\sin n\pi x|\, dx + \int_0^{\alpha} 2\,|\sin n\pi x|\, dx$$

$$= 4\frac{k}{n\pi} + \frac{2(1-\cos n\pi\alpha)}{n\pi}$$

$$\frac{h(n)}{a} = \frac{4(a-\alpha)}{a\pi} + \frac{2(1-\cos n\pi\alpha)}{n\pi}$$

$$= \frac{4}{\pi} + \frac{2(1-\cos n\pi\alpha)}{n\pi a} - \frac{4\alpha}{a\pi}$$

$\frac{h(n)}{a}$ is independent of n if $\alpha = 0$.

i.e. $\quad an = k \Rightarrow a = kN$

a should be divisible by 2, 3, 4 i.e. minimum (a) = 12.

14. $h(1) = \int_0^a \sin^2 x. \,|\cos x|\, dx$

$$\Rightarrow \lim_{a\to\infty} \frac{h(1)}{a} = \lim_{a\to\infty} \frac{1}{a}\int_0^a \sin^2 x. \,|\cos x|\, dx$$

Let $\quad a = n\pi + \alpha \quad 0 \le \alpha < \pi$

$$\Rightarrow \lim_{n\to\infty} \int_0^{n\pi+\alpha} \sin^2 x. \,|\cos x|\, dx = \lim_{n\to\infty} . \frac{1}{n\pi+\alpha}\left(n\int_0^{\pi} \sin^2 x\,|\cos x|\, dx + \int_0^{\alpha} \sin^2 x. \,|\cos x|\, dx \right)$$

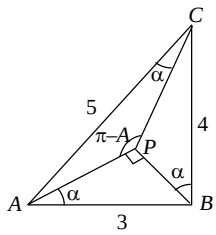
$$= \frac{1}{\pi}\int_0^{\pi} \sin^2 x\,|\cos x|\, dx = \frac{2}{3\pi}$$

15. $AP = 3\cos\alpha$

In ΔAPC

$$\frac{3\cos\alpha}{\sin\alpha} = \frac{5}{\sin(\pi - A)}$$

$$\Rightarrow \cot\alpha = \frac{25}{12}$$



16. $$\frac{\Delta PBC}{\Delta PAB} = \frac{\frac{1}{2}.PB.4.\sin\alpha}{\frac{1}{2}.PB.3.\sin(90^\circ - \alpha)} = \frac{4}{3}.\tan\alpha$$

$$= \frac{4}{3}.\frac{12}{25} = \frac{16}{25}$$

17. Either the stripes at odd number are painted with basic colour or at the even numbers.

Total ways $= 3^{\frac{n+1}{2}}\ 4^{\frac{n-1}{2}} + 3^{\frac{n-1}{2}}\ 4^{\frac{n+1}{2}} \qquad = 7(12)^{\frac{n-1}{2}}$

18. $t_n \rightarrow$ No of ways of painting *n* stripes

$a_n \rightarrow$ No of way of painting *n* stripes when it ends with a particular basic colour (*R*, *B* or *G*)

$b_n \rightarrow$ No of way of painting *n* stripes when it ends with a particular non basic colour (R/B, B/G or R/G)

$t_n = 3(a_n + b_n)$ (i)

$a_n = 2a_{n-1} + b_{n-1}$ (ii)

$b_n = a_{n-1}$ (iii)

Equation (ii) and (iii) givens

$$b_{n+1} = 2b_n + b_{n-1}$$

On solving characteristic equation $x^2 - 2x - 1 = 0$

we get $b_n = \alpha\left(1+\sqrt{2}\right)^{n-1} + \beta\left(1-\sqrt{2}\right)^{n-1}$

$b_2 = 1 \quad \Rightarrow 1 = \alpha + \beta + \sqrt{2}(\alpha - \beta)$

$\Rightarrow \alpha = \beta = \frac{1}{2} \quad \Rightarrow b_n = \frac{1}{2}\left(\left(1+\sqrt{2}\right)^{n-1} + \left(1-\sqrt{2}\right)^{n-1}\right)$

$$t_n = \frac{3}{\sqrt{2}}\left(\left(1+\sqrt{2}\right)^n - \left(1-\sqrt{2}\right)^n\right).$$

SECTION – C

1. Required plane contains z-axis and normal of $3x - 4y + z + 1 = 0$ hence its equation is $4x + 3y = 0$ distance from (1, 2, 3) is $\frac{4+6}{\sqrt{16+9}} = 2$ units.

2. On adding and subtracting the curves we get

$(x+3)^2 = 4(y-1),\ y^2 - 20y + 59 = 0$ respectively

Let point of intersection be $(x_1, y_1)(x_2, y_1)(x_3, y_2)(x_4, y_2)$

$\Rightarrow \quad y_1 + y_2 = 20$ and $y_1, y_2 > 0$

We required $\sum\sqrt{(x_1+3)^2+(y_1-2)^2} = \sum\sqrt{4(y_1-1)+(y_1-2)^2}$

$= \sum y_1 = 2(y_1+y_2) = 40$

3. $A = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 2 & 0 & 0 & \cdots & 0 \\ 3 & 0 & 0 & \cdots & \\ \vdots & & & & \\ n & 0 & 0 & \cdots & 0 \end{bmatrix}\begin{bmatrix} 2^{-1} & 2^{-2} & 2^{-3} & \cdots & 2^{-n} \\ 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ \vdots & & & & \\ 0 & 0 & 0 & \cdots & 0 \end{bmatrix}$

$A^2 = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 2 & 0 & 0 & \cdots & 0 \\ 3 & 0 & 0 & \cdots & \\ \vdots & & & & \\ n & 0 & 0 & \cdots & 0 \end{bmatrix}\begin{bmatrix} 2^{-1} & 2^{-2} & 2^{-3} & \cdots & 2^{-n} \\ 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ \vdots & & & & \\ 0 & 0 & 0 & \cdots & 0 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 2 & 0 & 0 & \cdots & 0 \\ 3 & 0 & 0 & \cdots & \\ \vdots & & & & \\ n & 0 & 0 & \cdots & 0 \end{bmatrix}\begin{bmatrix} 2^{-1} & 2^{-2} & 2^{-3} & \cdots & 2^{-n} \\ 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ \vdots & & & & \\ 0 & 0 & 0 & \cdots & 0 \end{bmatrix}$

$A^2 = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 2 & 0 & 0 & \cdots & 0 \\ 3 & 0 & 0 & \cdots & \\ \vdots & & & & \\ n & 0 & 0 & \cdots & 0 \end{bmatrix}\begin{bmatrix} \sum i2^{-i} & 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & 0 & \cdots & 0 \\ 0 & & & & & \\ \vdots & & & & & \\ 0 & 0 & 0 & 0 & \cdots & 0 \end{bmatrix}\begin{bmatrix} 2^{-1} & 2^{-2} & 2^{-3} & \cdots & 2^{-n} \\ 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ \vdots & & & & \\ 0 & 0 & 0 & \cdots & 0 \end{bmatrix}$

$\Rightarrow \quad A^2 = \left(\sum i.2^{-i}\right)A$

$\Rightarrow \quad A^n = \left(\sum i.2^{-i}\right)^{n-1}$

$\Rightarrow \quad Trace\, A^n = \left(\sum i.2^{-i}\right)^{n-1}\left(\sum a_{ii}\right) = \left(\sum i.2^{-i}\right)^n$

$\Rightarrow \quad \lim_{n\to\infty}\left(Trace\, A^n\right)^{1/n} = \lim_{n\to\infty}\left(\sum i2^{-i}\right) = \frac{1}{2}+\frac{2}{4}+\frac{3}{8}+\ldots.. = 2$

4. $\sum_{x=1}^{1985} f(x+1) = \sum_{x=1}^{1985}(-1)^{x+1}x - 2\sum_{x=1}^{1985} f(x)$

$\Rightarrow$ As $f(1) = f(1986)$

$\Rightarrow 3\sum_{x=1}^{1985} f(x) = \sum_{x=1}^{1985}(-1)^{x+1}x$

$\Rightarrow \sum_{x=1}^{1985} f(x) = 331.$

Sum of the digits = $3+3+1=7$.

5. From the figure

$\theta = 3\alpha = 129$

$8\cos^2(120^o) = 2$

